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Experimental and constitutive modelling studies of semicrystalline thermoplastics under solid-state stamp forming conditions

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ABSTRACT

Experimental characterisation and constitutive modelling studies on the thermomechanical behaviour of thermoplastics, under solid-state stamp forming conditions, are required for understanding and optimising the stamp forming process. In this paper, two semicrystalline thermoplastics, Polyamide 6 (PA6, or Nylon 6) and Poly-Ether-Ether-Ketone (PEEK) are studied via uniaxial tensile tests at temperatures between their glass transition temperatures (T_g) and melting temperatures (T_m), and at different strain rates (0.001–50 /s). The temperature and strain rate effects are analysed quantitatively to further understand the thermomechanical response of these semicrystalline thermoplastics. The results show that temperature has significant effects on the thermomechanical behaviour of thermoplastic polymers, while the strain rate effects are relatively marginal in the investigated strain rate range. In addition, a new physically-based viscoelastic-viscoplastic constitutive model is proposed to simulate the thermomechanical behaviour of both materials. The model shows good prediction accuracy on tensile stress responses and provides an insight into the microstructural evolution of the semicrystalline thermoplastics; thus, it can be used to analyse solid-state stamp forming of pure semicrystalline thermoplastics and thermoplastic polymer matrix composites (TPMCs).

1. Introduction

Polyamide 6 (PA6, or Nylon 6) and Poly-Ether-Ether-Ketone (PEEK) are two types of preferred thermoplastics in the automotive and aerospace industries, respectively [1,2]. They can serve as either pure polymers or as matrices for thermoplastic polymer matrix composite (TPMC) components. Both applications have been attracting increasing research and industrial interest, especially in relation to their manufacturing technologies. Traditional manufacturing technologies for sheet thermoplastics and TPMC components, such as compression moulding or autoclave forming, normally process material at melt state. These technologies require a long cycle time and involve high costs. In contrast, the stamp forming of thermoplastics and TPMCs at solid state can reduce the cycle time and improve the flexibility of the manufacturing process [2]. Since the polymer matrices contribute the main deformation constituent during forming, investigating the thermomechanical properties of thermoplastics under solid-state stamp forming conditions is vitally important for further study of the stamp forming process of thermoplastics and TPMCs.

The thermomechanical properties of thermoplastics are usually characterised by tensile or compression tests at various test conditions. Particular attention is paid to tensile properties for stamp forming applications [3]. Brinson et al. [4] summarised typical stress-strain curves of thermoplastics, and found that they are nonlinear responses and are combinations of viscoelastic and viscoplastic behaviours. These viscoelastic and viscoplastic behaviours reflect the time, strain rate, temperature, and pressure dependencies of thermoplastics, and of these parameters, the temperature and strain rate are the two most dominant parameters and have attracted greater attention from researchers [5]. For PA6, there are only a few studies on its thermomechanical properties. Tensile tests at a range of temperatures (23, 40, 60, 80, and 120 °C) were conducted by Parodi et al. [6] at a strain rate range from 10^{-4} to 10^{-1}/s. It was found that the tensile stress-strain curves changed noticeably with the temperature increase from room temperature to an elevated temperature. With increasing temperature, the elastic modulus and yield stress decreased, while the ductility increased. Further, at lower temperatures (23 and 40 °C), there was strain softening followed by strain hardening at large deformation, and the strain softening behaviour gradually disappeared at higher temperatures (60, 80, and 120 °C), at which only strain hardening was observed. The strain rate effects on the stress responses, however, were not significant within the

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Nomenclature

T_g, T_m	The glass transition temperature, the melting temperature (Celsius temperature)
ρ, ρ_c, ρ_a	The densities of the thermoplastic, the pure crystal phase, and the pure amorphous phase
χ_c	The degree of crystallinity
ε_e, ε_t	The experimentally-measured engineering strain and true strain
σ_e , σ_t	The experimentally-measured engineering stress and true stress
p_1, p_3	The positions of the 1st and 3rd marked lines along elongation throughout the test
l_0	The initial distance between the 3rd and 1st marked lines
w_0, t_0	The initial width and initial thickness at the gauge section of the tensile specimen
F	The load measured during the tensile test
$\boldsymbol{\Omega}_0, \overline{\boldsymbol{\Omega}}, \overline{\overline{\boldsymbol{\Omega}}}, \breve{\boldsymbol{\Omega}}, \boldsymbol{\Omega}$	The reference, intermedia plastic, intermolecular intermedia relaxed, network intermedia relaxed, and current configurations
$\boldsymbol{F}$, $\boldsymbol{F_I}$, $\boldsymbol{F_N}$	Deformation gradient, intermolecular deformation gradient, network deformation gradient
$\boldsymbol{F_{Ie}}$, $\boldsymbol{F_{Iv}}$, $\boldsymbol{F_{Ip}}$	The intermolecular elastic, viscous, and plastic deformation gradients
$\boldsymbol{F_{Ne}}$, $\boldsymbol{F_{Nv}}$	The network elastic and viscous deformation gradients
∇_X	The material gradient
$\boldsymbol{B_{Ie}}$	The intermedia elastic left Cauchy-Green deformation tensor
θ, θ_{ref}, θ_g, θ_m	The current, reference, glass transition, and melting temperatures (absolute temperature)
$\boldsymbol{\sigma}$	The Cauchy (true) stress tensor
$\boldsymbol{\sigma_I}$, $\boldsymbol{\sigma_N}$	The stress tensors of intermolecular and the network resistance
ε_{Ie}, ε_{Iv}, ε_{Ip}	The effective (true) strains of the intermolecular elastic, viscous, and plastic parts
ε_{Ne}, ε_{Nv}	The effective (true) strains of the network elastic and viscous parts
$\dot{\varepsilon}$	The total effective (true) strain rate
$\dot{\varepsilon}_{Ie}$, $\dot{\varepsilon}_{Iv}$, $\dot{\varepsilon}_{Ip}$	The effective (true) strain rates of the intermolecular elastic, viscous and plastic parts
$\dot{\varepsilon}_{Ne}$, $\dot{\varepsilon}_{Nv}$	The effective (true) strain rates of the network elastic and viscous parts
C_1	The temperature-dependent parameter in the intermolecular resistance
E	The temperature-dependent elastic modulus
E_{ref}	The temperature-dependent elastic modulus at the reference temperature
E_1, m_E	The material constants relating to E
ν	The Poisson's ratio
λ_{Ie}	The intermolecular elastic stretch ratio
σ_{lim}	The stress limit set on the viscoplastic friction element
σ_{ref}, m_σ	The material constants relating to σ_{lim}
$\dot{\gamma}_{01}$, m_{rI}, m_{Iv}	The material constants relating to the dashpot in the intermolecular viscoelastic part
η_{Iv}	The coefficient of viscosity of the dashpot in the intermolecular viscoelastic part
η_{Iv0}, $m_{\theta I}$	The material constants relating to η_{Iv}
σ_{Iv}, σ_{Ip}, σ_{Nv}	The effective stresses of the intermolecular viscous, intermolecular plastic and network visous parts
$\dot{\varepsilon}_{0p}$, m_{Ip}	The material constants relating to the dashpot in the viscoplastic part
C_p	The temperature-dependent parameter relating to the dashpot in the viscoplastic part
C_{p1}, m_{cp}	The material constants relating to C_p
C_R	The temperature-dependent modulus of the network resistance
C_{Rref}, C_{r1}	The material constants relating to C_R
J_{Ne}	The Jacobian determinant associated with the network elastic deformation gradient
$\boldsymbol{B}^*_{Ne}$	The deviatoric network elastic left Cauchy-Green deformation tensor
$\boldsymbol{F}^*_{Ne}$	The deviatoric part of $\boldsymbol{F_{Ne}}$
$\boldsymbol{I}$	The identity matrix
$\overline{\lambda}_{Ne}$	The average network elastic stretch ratio
$\overline{\lambda}_L$	The locking stretch
$\mathscr{L}^{-1}$	The inverse of the Langevin function
$\dot{\gamma}_{02}$, m_{rN}, m_{Nv}	The material constants relating to the dashpot in the network resistance
η_{Nv}	The coefficient of viscosity of the dashpot in the network resistance
η_{Nv0}, $m_{\theta N}$	The material constants relating to η_{Nv}
σ	The total effective stress
σ_c	The corrected total effective stress considering thermal softening by adiabatic heat
ω	The ratio of energy that is converted to temperature change to the total plastic dissipation
m_h	The material constant representing the softening parameter
θ_c	The corrected temperature during the deformation when the adiabatic heat comes into effect (absolute temperature)
$\dot{\theta}$	The temperature evolution (absolute temperature)
$\overline{C}$	The temperature-dependent specific heat capacity per unit volume
$\overline{C}_{ref}$, $\overline{C}_1$	The material constants relating to $\overline{C}$

studied conditions. For PEEK, the thermomechanical properties have been more extensively studied. Rae et al. [7] investigated the temperature effects on the tensile properties at temperatures between 50 – 150 °C, and a strain rate of 1.7 x 10^{-4} /s. With increasing temperature, the ductility increased remarkably while the yield stress and elastic modulus dropped, and this was especially significant when the temperature was above the glass transition temperature, T_g. In addition, their study showed that there was strain softening after material yielding at temperatures below 100 °C and no strain hardening behaviour was observed; whereas at higher temperatures (140, 150 °C), it was found that the strain softening behaviour disappeared and the strain hardening behaviour appeared. Zheng et al. [5] studied the temperature effects on PEEK at a temperature range between T_g and the melting temperature, T_m, and found that there was no obvious yield point at temperatures above T_g. However, their study only presented a limited strain level of the deformation. The strain rate effects of PEEK at room temperature were systematically studied by Alberola at al. [8] via tensile tests at a strain rate range from 10^{-5} to 300 /s. The strain rate effects on the stress responses were found to be marginal at the strain rate range from 10^{-5} to 10^{-1} /s. At higher strain rates (10^{-1} -300 /s), however, it was reported that there were appreciable increases in yield stress. The strain rate effects at elevated temperatures were studied by Chen et al. [9], and they found that there were no obvious strain rate effects at temperatures of 100, 140, and 150 °C. The strain rate effects for PEEK above T_g were studied at 240 °C at strain rates ranging from 10^{-3} to 10^{-1} /s by Barba et al. [10]. Although the strain rate effects were not significant at the

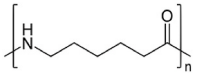
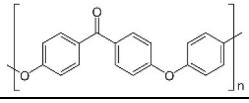
studied range, there were nonnegligible adiabatic heat effects. To summarise, there are studies that have investigated the thermomechanical properties of PA6 and PEEK polymers at various conditions. However, the material behaviour between the T_g and T_m (the T_g and T_m are respectively 47 °C and 230 °C for PA6, and 143 °C and 343 °C for PEEK), where the forming temperature is normally set to obtain the optimal formability for solid-state stamp forming [2], is still unclear, especially when the temperature is approaching T_m. Therefore, there is demand for comprehensive studies on the tensile behaviour between the T_g and T_m.

Besides the experimental studies, considerable attention has also been paid to the constitutive models of thermoplastic polymers, which are essential for numerical simulation on the stamp forming process. There are two approaches to establish constitutive models: the phenomenological approach, based on the macroscopic observation of the stress-strain response, and the physical approach, based on the physical interactions of the microstructures. Many phenomenological models are adopted from models developed for metal forming, such as the Johnson-Cook (JC) model. For example, Garcia-Gonzalez et al. [11] adopted the JC model to represent the viscoplastic behaviour of PEEK while simulating PEEK components under impact. Chen et al. [9] proposed a modified JC model based on mechanical test results to improve the JC model prediction under various temperatures and strain rates. However, the phenomenological models are generally used for engineering applications where the strain level is limited, as these models are not capable of capturing the viscoelastic and viscoplastic behaviours of thermoplastics at large deformation [12].

The physically-based models are built based on multiple sources of entropic force of molecular movement [11]: the separation, reorientation, and stretching of the molecular chains. From the microstructural point of view, both PA6 and PEEK are semicrystalline thermoplastics comprising well-arranged crystals and amorphous molecular chains. However, many existing physical models are built for amorphous polymers, such as the Boyce-Arruda model [13,14], based on an eight-chain assumption, and applied to amorphous thermoplastics. Xiao et al. [15] employed a back stress tensor for molecular orientations to improve the strain hardening prediction capability for amorphous polymers. To understand the macro-yield process in amorphous polymers, Jiang et al. [16] established the constitutive model via studying two molecular chain microstructures, i.e., sub-entanglement and topological entanglement. In their following study [17] on polymethyl methacrylate (PMMA), they illustrated that these two microstructures could reasonably be used to interpret the rate- and temperature-dependent macro-yield behaviour of amorphous polymers. Compared to amorphous polymers, the microstructural interactions are more complicated in semicrystalline thermoplastics since there are not only interactions between multiple microstructure phases, but also phase transformation during deformation [10]. Some studies on the physically-based models for semicrystalline thermoplastics have been conducted in recent years. For example, to describe the stress responses of semicrystalline thermoplastics at room temperature, Polanco-Loria et al. [18] built a hyperelastic-viscoplastic constitutive model by considering the pressure sensitivity, volume plastic strain, and strain rate sensitivity under isothermal conditions. Abdul-Hameed et al. [19] made progress by revising Boyce et al. [14]'s model, combining the influence of degree of crystallinity to study the strain rate effects on semicrystalline thermoplastics. Berezvai et al. [20] further studied the strain rate effects on thermoplastics by establishing a constitutive model based on the contributions of crystalline, amorphous, and rigid amorphous fractions. To understand the thermal effects, Garcia-Gonzalez et al. [5] improved Polanco-Loria's model [18] with consideration of thermal-mechanical coupling, including the effects of thermal softening, thermal expansion, and the adiabatic heat caused by plastic dissipation during deformation. For considering the material behaviour of thermoplastics at temperatures above T_g, Barba et al. [10] studied a broader application of the model of Garcia-Gonzalez et al. [5] with experimental data on PEEK corresponding to three thermal regimes, which were below, at, and above T_g, so that the reorientation mechanism of the crystal phase and the amorphous phase could be better understood. However, the model predictions showed noticeable discrepancies compared with experimental data. To conclude, the constitutive models aiming to predict the thermomechanical behaviour at temperatures between T_g and T_m are still not well studied. Therefore, there is demand for a physically-based constitutive model with comprehensive consideration of temperature and strain rate effects to accurately predict the thermomechanical behaviour.

The objective of this paper is to comprehensively characterise the thermomechanical properties of two typical semicrystalline thermoplastics, under solid-state stamp forming conditions, via experimental studies and constitutive model development. In this research, uniaxial tensile tests were performed on PA6 and PEEK, under a temperature range between T_g and T_m (roughly, 0.7–0.9 T_m at absolute temperature) and a strain rate range from 0.001 to 50 /s, which were defined corresponding to typical solid-state stamp forming conditions. A new physically-based viscoelastic-viscoplastic constitutive model with comprehensive consideration of the temperature and strain rate effects was proposed. The model parameters applied to PA6 and PEEK were determined based on the characterised materials' properties. The developed constitutive model was subsequently employed to predict the stress-strain behaviour of these materials at different temperatures and strain rates for stamp forming applications.

Table 1
Mechanical and thermal properties of PA6 and PEEK.

Properties	PA6	PEEK
Elastic modulus at 20°C (GPa)	3.5	4.3
Yield stress at 20°C (MPa)	80	115
Elongation at yield at 20 °C (%)	4	5
Elongation at break at 20°C (%)	55	7
Density (kg/m^3)	1150	1310
Density of crystal phase and amorphous phase (kg/m^3)	ρ_c = 1210 ρ_a = 1090	ρ_c = 1400 ρ_a = 1260
Glass transition temperature T_g (°C)	47	143
Melting temperature T_m (°C)	230	343
Coefficient of thermal expansion (/K)	80 – 90 × 10^{-6}	50 – 70 × 10^{-6}
Molecular weight (g/mol)	113.16	288.31
Repeat unit		

2. Experimental methods

2.1. Material and specimen design

The PA6 and PEEK materials were supplied by Direct Plastics and Ai Engineering Plastics & Laminates, respectively, in the UK. The mechanical and thermal properties of the investigated PA6 and PEEK are provided in Table 1.

Since both the PA6 and PEEK polymers are semicrystalline thermoplastics, the degree of crystallinity in mass fraction, χ_c, can be estimated according to Refs. [11,21,22]:

$$\chi_c = \frac{\rho_c(\rho - \rho_a)}{\rho(\rho_c - \rho_a)} \quad (1)$$

where, ρ is the density of the thermoplastic and ρ_c, ρ_a are the densities of the pure crystal phase and the pure amorphous phase, respectively. The degrees of crystallinity of PA6 and PEEK are estimated to be approximately 52% and 30%, respectively.

The uniaxial tensile tests were conducted based on ASTM 638 [23] which is a standard tensile testing method for plastics. Considering the large elongation of thermoplastics at elevated temperatures, the dimensions of the tensile test specimen design were determined and are shown in Fig. 1a. Two holes were added on the specimen at the two ends for mounting onto the grips. The gauge length was defined as 7.62 mm according to the testing standard. The thicknesses of PA6 and PEEK were 2 mm and 5.5 mm, respectively. The PA6 and PEEK specimens were manufactured by laser cutting and water jet cutting, respectively, and the dimensions were inspected. The manufactured tensile test specimens of PA6 and PEEK are shown in Fig. 1b and c, respectively.

2.2. Experimental setup

For both types of thermoplastics, the tested temperature range was around 0.7–0.9 T_m on the absolute temperature scale, in which range the solid-state polymer materials are more deformable compared to room temperature; the tested strain rate range was set at 0.001–50 /s which covers stamp forming conditions [24]. The test conditions for PA6 and PEEK are listed in Table 2. For each condition, tests were repeated three or four times to check their repeatability.

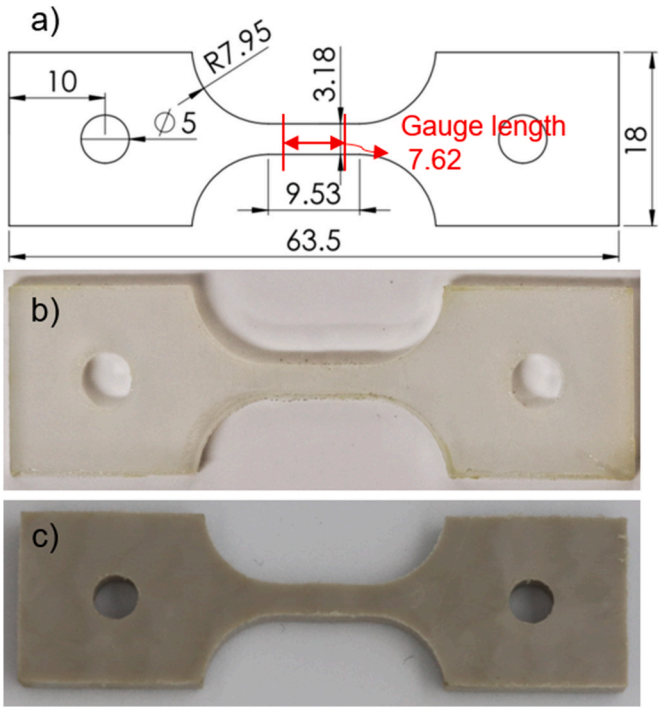


Fig. 1. Uniaxial tensile test specimens: a) the design and dimensions; photographs of b) PA6 and c) PEEK tensile test specimens. (unit: mm).

Table 2
Uniaxial tensile test matrices.

Test equipment	Low-rate INSTRON			High-rate INSTRON		
Strain rate (/s) / Temperature (°C)	0.001	0.01	0.1	1	10	50
PA6						
160			✓			
180		✓	✓	✓	✓	
200			✓			
220		✓	✓	✓	✓	
PEEK						
180			✓			
210			✓			
240	✓	✓	✓	✓	✓	✓
270			✓			
300			✓			
310			✓			

Fig. 2 shows two sets of experimental setups for a) low-rate and b) high-rate tests, respectively. All of the low-rate tests were performed on an INSTRON 5584 tensile testing machine, shown in Fig. 2a, and all of the high-rate tests were performed on an INSTRON VHS servo-hydraulic machine, shown in Fig. 2b. An environmental chamber was mounted to each machine. The average heating rate was around 4 °C/min. The tested temperature on the surface of the specimens was monitored by using thermocouples connected to a data-logger. To achieve a uniform temperature distribution, the specimens were soaked at testing temperatures for 10 min before the tests started. Fans were used to prevent the load cell and the machine from overheating. A 60 frames-per-second (fps) low-speed camera and a 500 fps high-speed camera were used to capture the deformation at low-rate and high-rate tests, respectively. A light was used on each setup to assist in capturing the deformation images. Bespoke lightweight tensile test grips were used to reduce the inertia effects during the high-rate tests.

2.3. In-situ measurement

In order to measure the extra-large tensile strains of the thermoplastics at elevated temperatures, an in-house Digital Image Correlation (DIC) script was developed in MATLAB. As shown in Fig. 3, the specimen was marked with three lines, where the 1st and 3rd lines marked the two ends of the gauge length (7.62 mm), and the 2nd line marked the middle of the gauge length. The script recognised and tracked the three marked lines by detecting the differences in pixel values along the specimen using a sliding window schematically marked in Fig. 3a. After detecting one frame of images, the differences in pixel values along the elongation could be obtained, shown as Fig. 3b. The three marked lines' pixel positions along the elongation were extracted by locating the pixel positions of the three peaks in Fig. 3b. By conducting the same process on all the frames, the pixel positions of the three marked lines along the elongation were obtained as shown in Fig. 3c. Therefore, the evolution of the average strain within the gauge length could be calculated from the distance change between the 1st and 3rd lines. As illustrated by the example in Fig. 3c, the 2nd line is always in the middle of the 1st and 3rd line, demonstrating a uniform deformation within the gauge length throughout the test. It is worth mentioning that not all test results were under uniform deformation until fracture. Necking behaviour was observed at several high-rate or low-temperature tests.

The true tensile strain and true tensile stress were calculated according to Eqs. (2)– (5).

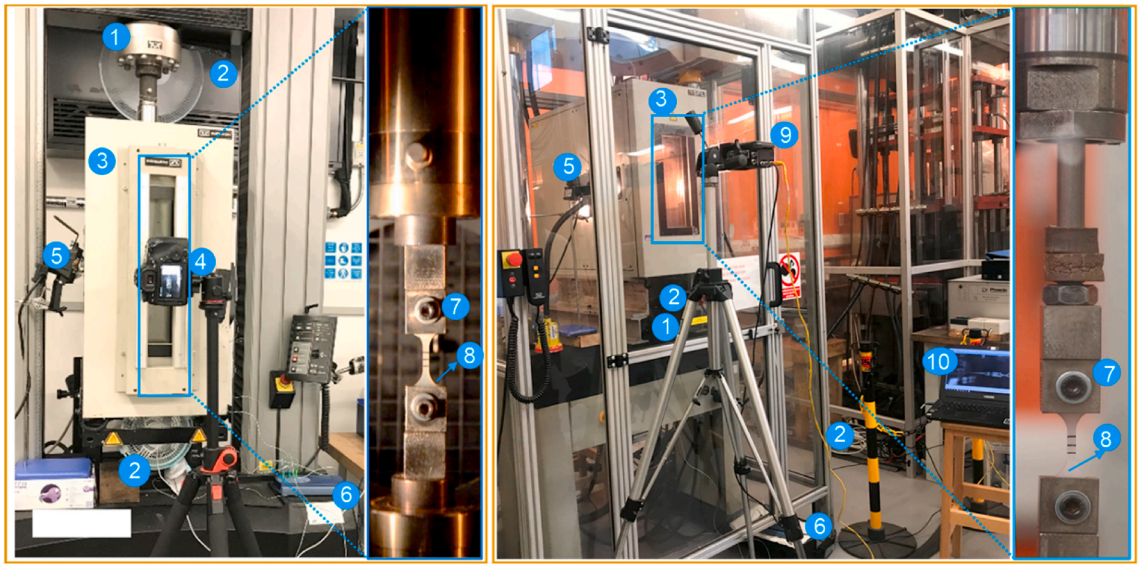


Fig. 2. Experimental setup. a) Test setup on low-rate INTRON; b) Test setup on high-rate INSTRON; ①Load cell; ②Fan; ③Environmental chamber; ④Low-speed camera; ⑤Light; ⑥Thermocouple and data-logger; ⑦Tensile test grip; ⑧Tensile test specimen; ⑨High-speed camera; ⑩ Laptop and software to capture the image from the high-speed camera.

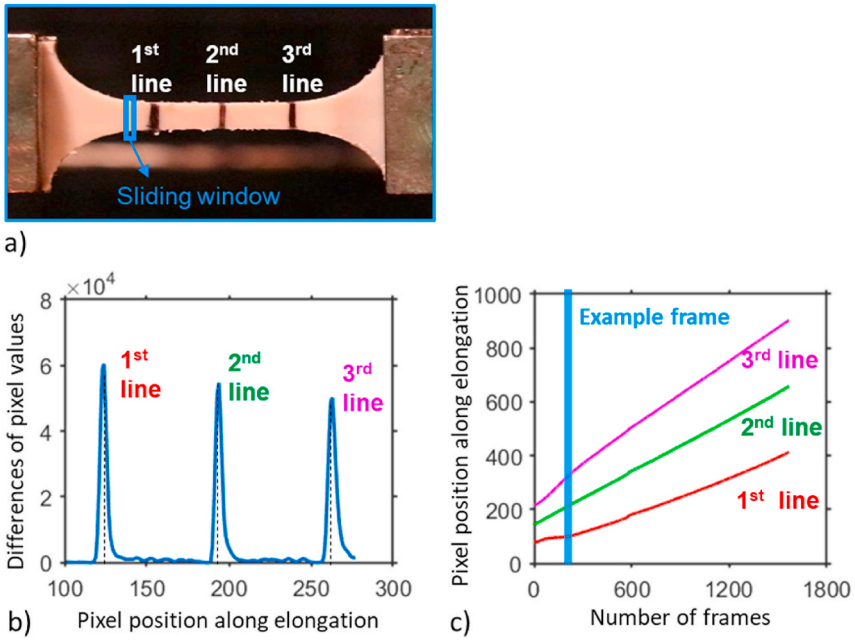


Fig. 3. Tensile strain captured using an image processing method. a) An example frame of the deformation pictures; b) Identifying the position of marked lines through pixel values in the example frame; c) The tracked positions of three marked lines throughout a uniaxial tensile test.

$$\varepsilon_e = \frac{p_3 - p_1 - l_0}{l_0} \tag{2}$$

$$\varepsilon_t = \ln(1 + \varepsilon_e) \tag{3}$$

$$\sigma_e = \frac{F}{w_0 t_0} \tag{4}$$

$$\sigma_t = \sigma_e(1 + \varepsilon_e) \tag{5}$$

where, ε_e and ε_t are the experimentally-measured engineering tensile strain, and true tensile strain, respectively. p_3, p_1 are the positions of the 3rd line and the 1st line along elongation throughout the test as shown in Fig. 3, and l_0 is the initial distance between the 3rd line and the 1st line (i.e., gauge length). σ_e and σ_t are the experimentally-measured engineering tensile stress and true tensile stress, respectively. F is the measured load from the load cell on the tensile test machine. w_0 and t_0 are the initial width and initial thickness, respectively, at the gauge section measured from each tested specimen.

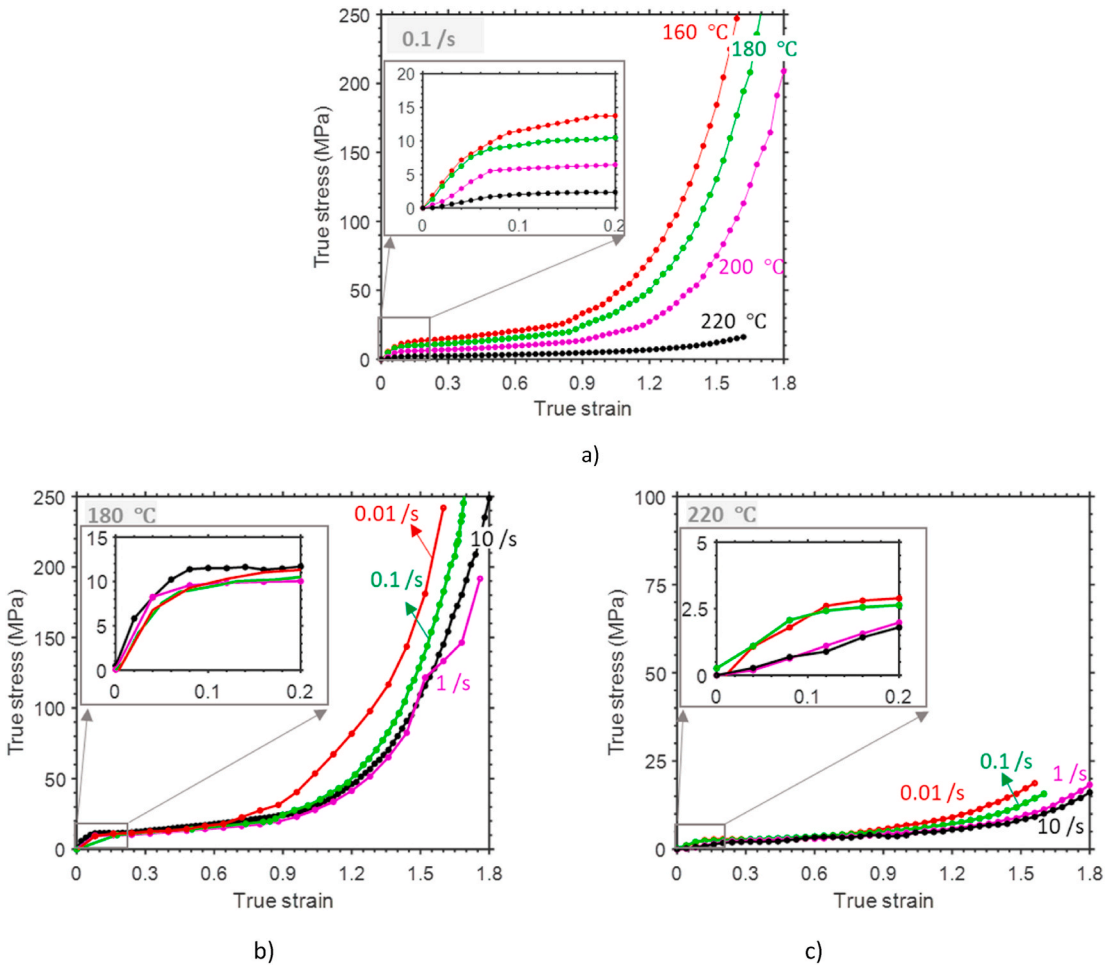


Fig. 4. Experimental true stress – true strain curves from uniaxial tensile tests for PA6 under different temperatures (160–220 °C) and strain rates (0.01–1 /s). a) Temperature effects at the strain rate of 0.1 /s; strain rate effects at temperatures of b) 180 °C and c) 220 °C.

3. Experimental results and analysis

3.1. Uniaxial tensile test results of PA6 and PEEK

Fig. 4 shows the uniaxial tensile test results of PA6. The true stress – true strain curves at different temperatures and the same strain rate of 0.1 /s are shown in Fig. 4a. There is a noticeable decrease in overall stress level with an increase in temperature. The initial elastic deformation part shows nonlinear properties, and no obvious yield points can be detected in the tested material. At larger deformation stages, there is a significant strain hardening behaviour when PA6 deforms at temperatures below or equal to 200 °C, whereas the strain hardening effect is

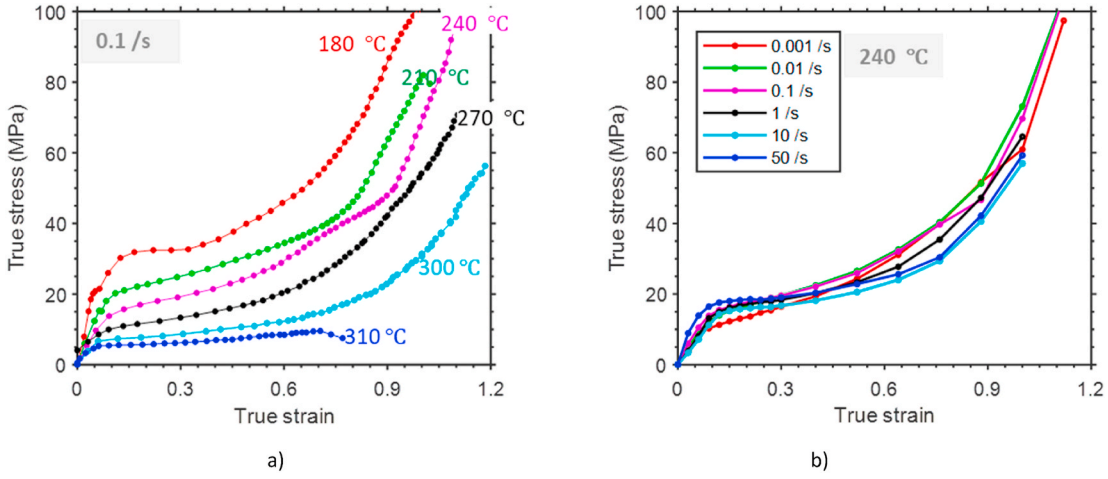


Fig. 5. Experimental true stress – true strain curves from uniaxial tensile tests for PEEK at different temperatures (180–310 °C) and strain rates (0.001–50 /s). a) Temperature effects at a strain rate of 0.1 /s. b) Strain rate effects at a temperature of 240 °C.

much reduced at 220 °C.

Fig. 4b and c show the tensile responses of PA6 under various strain rates at 180 °C and 220 °C, respectively. In general, for both temperatures, it can be observed that the strain rate effects, within the tested range of 0.01–10 /s, are not as obvious as the observed temperature effects. There is no significant difference in stress responses when the true tensile strain is less than 0.6. The effect of the strain rate in stress response at a larger tensile strain becomes more obvious. Interestingly, it is found that the stresses are higher at lower strain rates (0.01, 0.1 /s) when the tensile strain is higher than 0.6. Please note that the vertical scale in Fig. 4c is different with that in Fig. 4a and b since the stresses are relatively small.

Fig. 5 shows the uniaxial tensile test results of PEEK. The true stress – true strain curves at different temperatures and the same strain rate of 0.1 /s are shown in Fig. 5a. Similar to the temperature effects on PA6, there is a considerable decrease in the overall stress level with increasing temperature. At the initial stage of deformation, the material also shows nonlinear behaviour within the studied temperature range. At 180 °C, there is a region showing slight strain softening after the initial elastic deformation, which is believed to be related to the reorientation of crystallites and the fragmentation of larger crystalline phases [25]. This behaviour disappears at higher temperatures. The strain hardening phenomena appear at larger stages of deformation, when the temperature is lower than 310 °C. At 310 °C, the strain hardening behaviour is much reduced and is not obvious.

The strain rate effects on PEEK can be inferred from Fig. 5b. In general, the stress responses of PEEK at different strain rates, at 240 °C, are similar compared with those of PA6 at different strain rates (see Fig. 4b and c. The tensile stress is found to be higher at a higher strain rate within a low tensile strain level, and the opposite trend is observed when the tensile strain is higher than 0.6.

3.2. Key features of the stress-strain curves

Based on the tensile test results from Figs. 4 and 5, in order to conduct a more in-depth analysis of the temperature and strain rate effects, three stress response regions are qualitatively defined for each true stress – true strain curve [3], as illustrated in Fig. 6a. Region I is the initial nonlinear elastic region where the stress increases elastically with the increasing strain; region II is a pre-strain hardening region where the stress increases steadily with increasing strain; region III is a post-strain hardening region where the stress increases exponentially with increasing strain. Key features for the three regions are further characterised and discussed.

For region I, the elastic modulus and yield stress are analysed. As has been mentioned, there was no obvious yield point observed for the thermoplastics at the tested temperatures. Thus, it is difficult to locate the elastic limit from the true stress – true strain curves. Therefore, the extrinsic yield point is used in this study, as mentioned by Brinson et al. [4]. As shown in Fig. 6b, the extrinsic yield point is captured by drawing the tangent lines from points on the stress-strain curve and locating the point when the intersection of the tangent line and the horizontal axis is

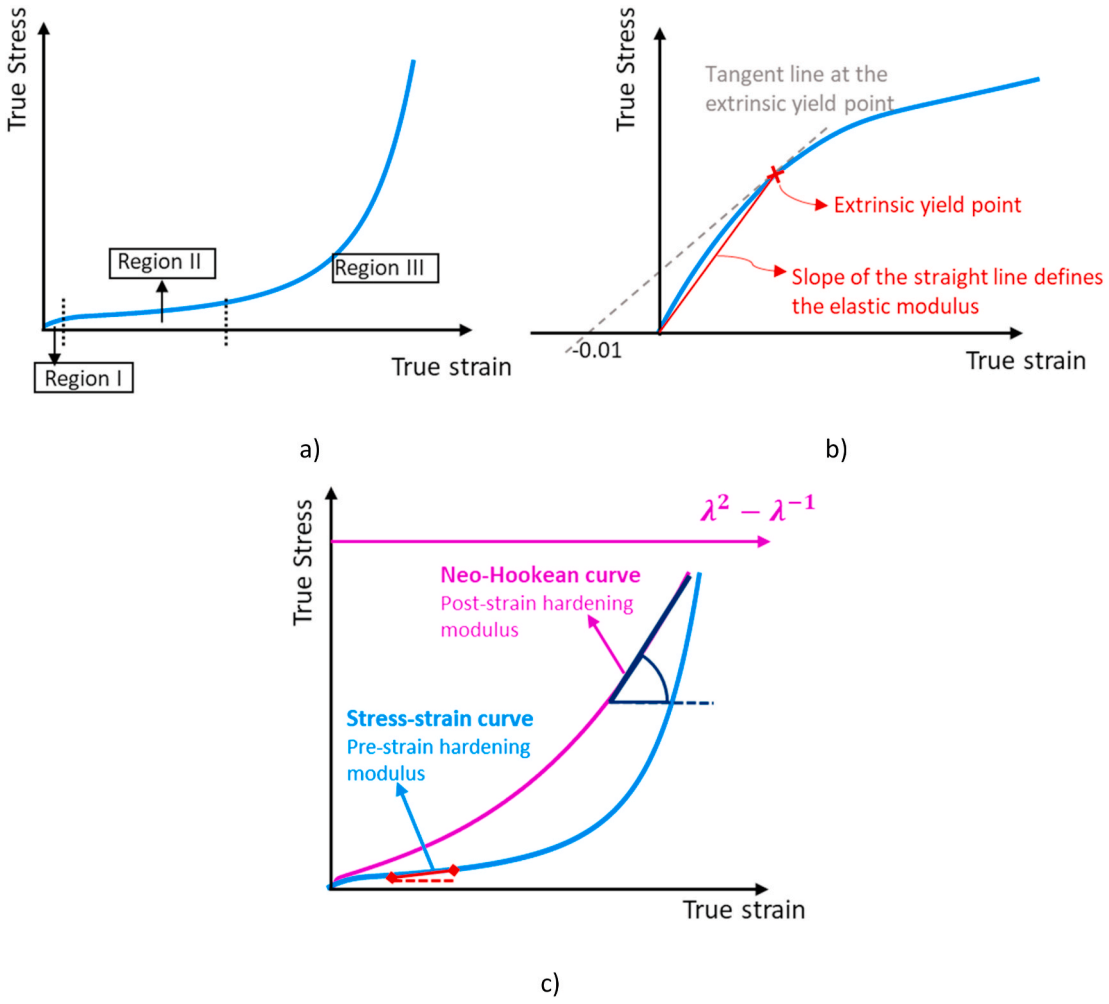


Fig. 6. The analysis of the key features of a representative stress-strain curve: a) three stress response regions; b) the identification of the yield point and elastic modulus; c) the definition of the pre-strain hardening modulus and the post-strain hardening modulus.

−0.01 [4]. The elastic modulus is defined by the slope of the line connecting the undeformed point (i.e. origin) and the extrinsic yield point.

In region II, the strain hardening behaviour appears. Although there is slight strain softening for PEEK at 180 °C, it lasts for only a short deformation stage and is also followed by a steady strain hardening stage. Therefore, Region II is identified as the pre-strain hardening stage and is described by a pre-strain hardening modulus. As can be observed from Figs. 4 and 5, at a strain range of 0.3–0.5, there is a steady strain hardening behaviour for both materials at all test conditions. Therefore, the pre-strain hardening modulus is calculated by the average gradient of the true stress – true strain curve at the strain range of 0.3–0.5, as shown in Fig. 6c.

For region III, defined as the post-strain hardening region, the stress response could be described by a Neo-Hookean curve according to Melick et al. [26], shown as the pink line in Fig. 6c. The Neo-Hookean curve is defined by the relationship between the true stress and the term $(\lambda^2 - \lambda^{-1})$, where $\lambda = \varepsilon_e + 1$ is the stretch ratio, representing the ratio between the final and the initial gauge length of the tested material. The post-strain hardening modulus is determined as the slope of the last portion of the Neo-Hookean curve, which is linear. In this study, the last 20% of the Neo-Hookean curve is used to calculate the post-strain hardening modulus.

3.3. Analysis of the temperature effects

The temperature effects on the stress response are analysed in detail as below.

1) The elastic modulus and the yield stress

The yield stresses and elastic moduli of PA6 and PEEK at different temperatures and the strain rate of 0.1 /s are shown in Fig. 7. It is clear that, with increasing temperature, the elastic modulus and the yield stress decrease nonlinearly for both materials.

2) The strain hardening behaviour

Fig. 8a shows the pre-strain hardening moduli of tensile deformation at different temperatures and a strain rate of 0.1 /s for both materials. As can be observed, with increasing temperature, the pre-strain hardening modulus decreases linearly. The extent of the temperature dependence of the pre-strain hardening modulus can be defined by the slope of each trendline, and the slopes for both materials are almost equal.

Fig. 8b shows the post-strain hardening moduli of tensile deformation at different temperatures and a strain rate of 0.1 /s for both materials. As can be observed, there are also linear relationships between the post-strain hardening modulus and the temperature for both materials in the lower temperature ranges (160–200 °C for PA6, 180–300 °C for PEEK). The slopes of the trendlines for PA6 and PEEK represent the temperature dependencies of post-strain hardening, and the slope for PA6 is higher than the slope for PEEK. At temperatures of 220 °C for PA6 and 310 °C for PEEK, the post-strain hardening moduli are near zero. This corresponds to Figs. 4a and 5a, where the post hardening disappeared at the temperatures of 220 °C and 310 °C for PA6 and PEEK, respectively. This phenomenon indicates that 220 °C and 310 °C are near the critical temperatures approaching T_m, for PA6 and PEEK, respectively.

3.4. Analysis of strain rate effects

From the tensile test results shown in Figs. 4b, c and Fig. 5b, the strain rate effects are not as obvious as the temperature effects. The elastic moduli, yield stresses, pre-strain hardening, and post-strain hardening moduli are almost the same at different strain rates for both PA6 and PEEK. However, there is a different trend at the early stage and the later stage of the deformation. Therefore, to demonstrate this phenomenon, the stress levels corresponding to two strain levels are analysed. Fig. 9 shows true stresses of PA6 tested at 180 °C and PEEK tested at 240 °C at the strain levels of 0.1 and 0.7, corresponding to region II and region III respectively, at various strain rates. It can be observed that, at a small strain level of 0.1, for PA6 at 180 °C or PEEK at 240 °C, the stress remains unchanged (for PA6) or increases (for PEEK) with increasing strain rate. On the contrary, at a larger strain level (i.e. 0.7), the stress slightly decreases with an increase in strain rate. The shift of strain rate effects on stress responses at different deformation stages indicates that the true stress level is not a monotonic function of strain rate for PA6 at 180 °C or PEEK at 240 °C, which can be attributed to the adiabatic heat at higher strain rates [27,28]. At higher strain rates, there is not sufficient time for the plastic deformation energy to dissipate. Therefore, fast deformation results in heat accumulation and temperature increase in the deformed area [10], which softens the material in the deformed area at the later stage of deformation. The existence of adiabatic heat can be proven by the necking behaviour observed in the tensile tests of PEEK, with an example shown in Fig. 10. In the tensile tests at various strain rates for PEEK at 240 °C, necking behaviour was found in the tests at the highest strain rate (50 /s), but not in a strain rate of 0.001 /s throughout the whole test. This adiabatic heat effect will be further discussed in the constitutive modelling part.

4. Constitutive model

4.1. Description of the model

To build up a physically-based viscoelastic-viscoplastic model for semicrystalline thermoplastics, rheological models are commonly used. The mechanical responses can be visually understood and expressed by

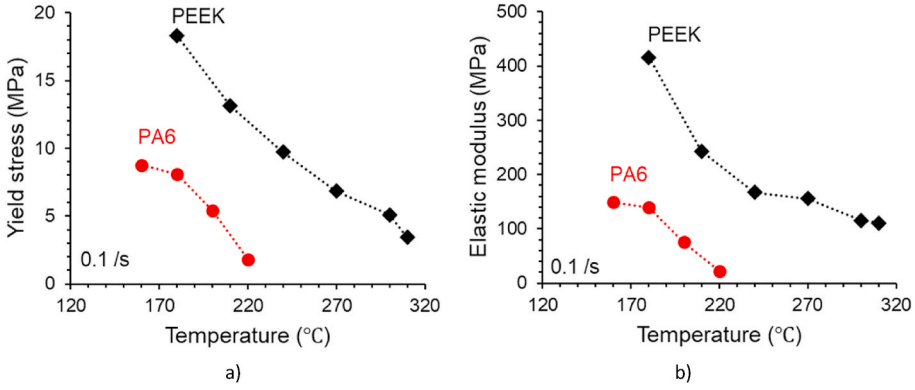


Fig. 7. Temperature effects on the a) yield stresses and b) elastic moduli of PA6 and PEEK at different temperatures (160–220 °C for PA6 and 180–310 °C for PEEK) and a strain rate of 0.1 /s.

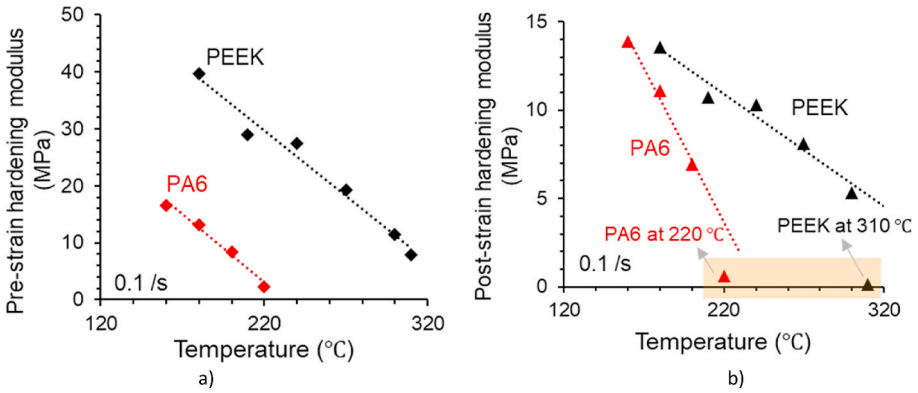


Fig. 8. Temperature effects on the a) pre-strain hardening modulus and b) post-strain hardening of PA6 and PEEK at different temperatures (160–220 °C for PA6 and 180–310 °C for PEEK) and a strain rate of 0.1 /s.

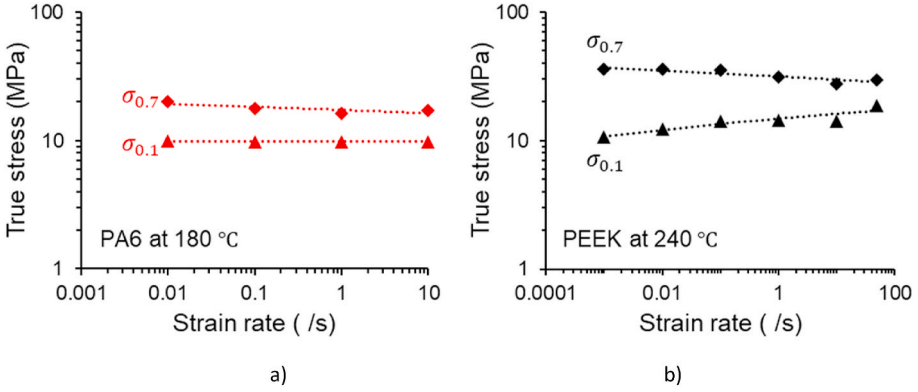


Fig. 9. Stress analysis at strain levels of 0.1 and 0.7 for a) PA6 at a temperature of 180 °C, and b) PEEK at a temperature of 240 °C, at various strain rates (0.001–50 /s). (The figures are in log-log scales).

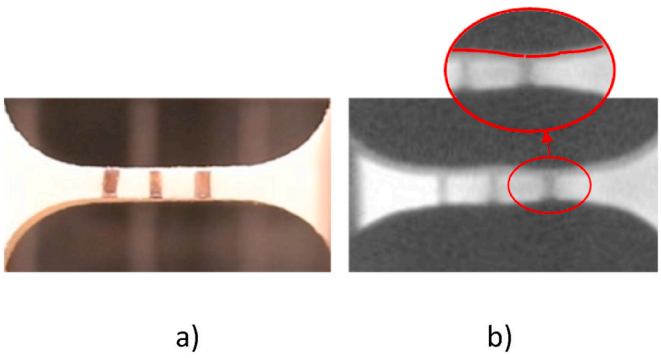


Fig. 10. An example of deformation images of PEEK at a temperature of 240 °C, and at a strain rate of a) 0.001 /s (no necking observed) and b) 50 /s (necking observed), at the same deformation stage ($\varepsilon_e = 0.14$).

combining simple responses of the basic elements, such as springs, dashpots, and friction elements, in the rheological models [29]. The stress-strain relationship can be derived according to the stress and strain generated in the rheological models. In this section, a new physically-based constitutive model is developed based on the rheological model proposed by Polanco-Loria [18]. In the Polanco-Loria model, there are two sources of resistance, i.e. the intermolecular resistance and the network resistance, to overcome for material deformation. These two sources of resistance are assembled in parallel with each other, which is also mentioned in other studies [13,30,31]. The Polanco-Loria model is detailed below:

- The intermolecular resistance (I). The intermolecular resistance is composed of an elastic part and a viscoplastic part in series. The elastic part is represented by a hyperelastic spring, while the viscoplastic part is represented by a dashpot and a friction element in parallel. The initial stiffness and flow stress are both results of the intermolecular interactions.
- The network resistance (N): The resistance against network stretching is represented by a hyperelastic spring. The strain hardening during deformation is related to the network resistance, which is generated from molecular reorientation and alignment.

In this study, based on the experimental characterisation and analysis of the tensile stress-strain response of PA6 and PEEK, a new constitutive model which further develops both the rheological model form and the mathematical expressions is proposed, aiming to predict the thermomechanical behaviour of semicrystalline thermoplastics under solid-state stamp forming conditions in an accurate manner. Fig. 11 shows the rheology scheme and the kinematics of the proposed model. Compared with the Polanco-Loria model, the new features in the rheological model form are that the hyperelastic springs in the intermolecular branch and the network branch are both replaced with a rate-dependent viscoelastic part, which is composed of a hyperelastic spring and a viscous dashpot [4]. In the mathematical expression of each rheological element, comprehensive temperature and strain rate effects are considered based on their physical meaning, which will be elaborated upon in detail in the following section.

Fig. 11a shows the rheology scheme of the proposed model, which is composed of an intermolecular resistance (I) in parallel with a network resistance (N). The intermolecular resistance (I) branch is composed of a

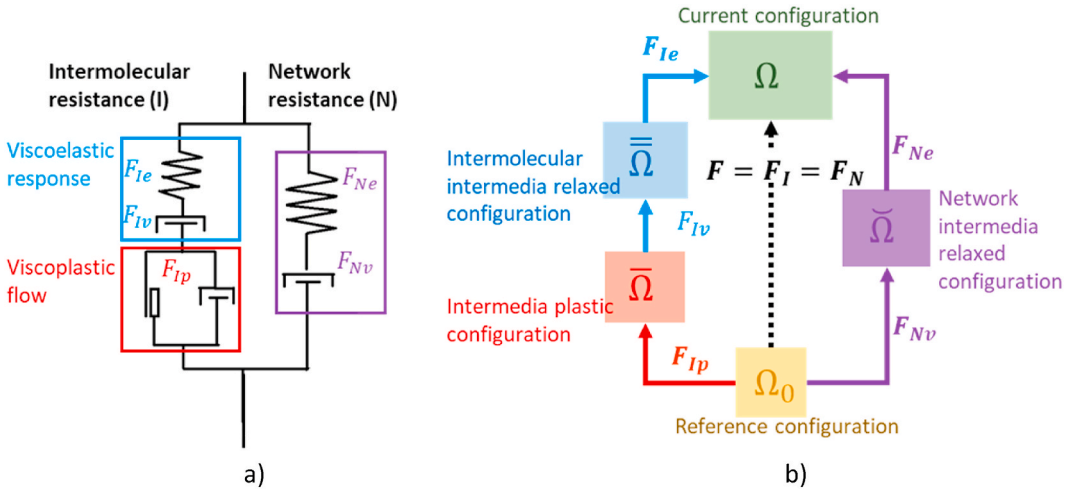


Fig. 11. The proposed new constitutive model. (a) The rheology scheme; (b) The kinematics ($\boldsymbol{F}$ is the deformation gradient, $\boldsymbol{F_I}, \boldsymbol{F_N}$ represent the intermolecular deformation and the network deformation gradients, respectively. $\boldsymbol{F_{Ie}}, \boldsymbol{F_{Iv}}, \boldsymbol{F_{Ip}}$ represent the intermolecular elastic, viscous, and plastic deformation gradients, respectively.$\boldsymbol{F_{Ne}}, \boldsymbol{F_{Nv}}$ represent the network elastic and viscous deformation gradients, respectively.).

viscoelastic part and a viscoplastic part in series. The viscoelastic deformation is represented by a temperature-dependent Neo-Hookean hyperelastic spring in series with a temperature- and rate-dependent dashpot [5,10,18]. The hyperelastic spring characterises the initial elastic deformation due to the internal energy change, and the dashpot in the viscoelastic part describes the viscosity contribution. The viscoplastic deformation is represented by a temperature- and rate-dependent dashpot in parallel with a friction element. It should be noted that the friction element is designed to represent the activation of plastic deformation, not as a physically microstructural response. When the stress in the friction element reaches the stress limit, the viscoplastic deformation begins. The stress contribution from viscoelastic deformation and viscoplastic deformation were analysed by Barba et al. [10]. Around 90% permanent deformation was found in PEEK at temperatures below and above T_g, which also shows good agreement with the present model (detailed later). The network resistance is composed of a temperature-dependent eight-chains hyperelastic spring [14] in series with a temperature- and rate-dependent dashpot. The network resistance (N) branch describes the strain hardening behaviour in the later deformation stage caused by the reorientation and alignment of molecular chains.

The kinematics analysis method of rheological models has been described in detail by Polanco-Loria et al. [18], Garcia-Gonzalez et al. [5,12], and Barba et al. [10]. Similarly, Fig. 11b shows the kinematics of this new model. Five configurations are established to define the deformation kinematics of the new model. The initial reference configuration, $\boldsymbol{\Omega_0}$, deforms via the combination of three intermedia configurations to the current configuration, $\boldsymbol{\Omega}$. The intermedia plastic configuration, $\bar{\boldsymbol{\Omega}}$, and the intermolecular intermedia relaxed configuration, $\bar{\bar{\boldsymbol{\Omega}}}$, account for the plastic deformation and the viscous deformation in the intermolecular resistance, respectively. The network intermedia relaxed configuration, $\breve{\boldsymbol{\Omega}}$,accounts for the viscous deformation in the network resistance.

Setting $\boldsymbol{X}$, $\boldsymbol{x}$ as an arbitrary material point in the reference configuration $\boldsymbol{\Omega_0}$ and the current configuration $\boldsymbol{\Omega}$, respectively, the mapping relationship can be represented as $\boldsymbol{x} = \chi(\boldsymbol{X},t)$, where t is time. Therefore, the deformation gradient, $\boldsymbol{F}$, can be derived as

$$\boldsymbol{F} = \nabla_X \chi \tag{6}$$

where ∇_X denotes the material gradient.

The deformation gradient, $\boldsymbol{F}$, is equivalent to both the intermolecular deformation gradient, $\boldsymbol{F_I}$, and the network deformation gradient, $\boldsymbol{F_N}$, according to the rheological model. Therefore,

$$\boldsymbol{F} = \boldsymbol{F_I} = \boldsymbol{F_N} \tag{7}$$

For the intermolecular side, the deformation gradient can be decomposed into the intermolecular elastic, $\boldsymbol{F_{Ie}}$, viscous $\boldsymbol{F_{Iv}}$, and plastic $\boldsymbol{F_{Ip}}$ deformation gradients. For the side associated with the network resistance, it is divided into network elastic $\boldsymbol{F_{Ne}}$, and viscous $\boldsymbol{F_{Nv}}$, deformation gradients. Therefore,

$$\boldsymbol{F_I} = \boldsymbol{F_{Ie}}\boldsymbol{F_{Iv}}\boldsymbol{F_{Ip}} \tag{8}$$

$$\boldsymbol{F_N} = \boldsymbol{F_{Ne}}\boldsymbol{F_{Nv}} \tag{9}$$

4.2. Development of a new viscoelastic-viscoplastic constitutive model

From the aforementioned rheology scheme, the Cauchy stress tensor (i.e. true stress tensor), $\boldsymbol{\sigma}$, in terms of the intermolecular resistance and the network resistance can be expressed as:

$$\boldsymbol{\sigma} = \boldsymbol{\sigma_I} + \boldsymbol{\sigma_N} \tag{10}$$

where, $\boldsymbol{\sigma_I}$ and $\boldsymbol{\sigma_N}$ denote the stresses contributed by the intermolecular resistance and the network resistance, respectively.

1) The intermolecular resistance (I)

The stress contributed by the intermolecular resistance is defined by a Neo-Hookean hyperelastic spring [32]. As Barba et al. [10] observed, polymers are nearly incompressible at temperatures above T_g; therefore, the incompressible Neo-Hookean material model is adopted. The Cauchy stress can be expressed as:

$$\boldsymbol{\sigma_I} = 2C_1(\theta)\boldsymbol{B_{Ie}} \tag{11}$$

where $\boldsymbol{B_{Ie}}$ is the intermolecular elastic left Cauchy-Green deformation tensor, calculated by $\boldsymbol{B_{Ie}} = \boldsymbol{F_{Ie}}\boldsymbol{F_{Ie}^T}$. With θ being the current temperature, the term $C_1(\theta)$ is a temperature-dependent parameter and is calculated by $C_1(\theta) = E(\theta)/(4(1+\nu))$, where ν is Poisson's ratio and $\nu = 0.48$ is used in this paper [10,32]. The term $E(\theta)$ is the temperature dependent elastic modulus. From the research of Bouvard et al. [33] and Garcia-Gonzalez et al. [5], the decrease in elastic modulus $E(\theta)$ was modelled as a linear relationship with increasing temperature. However, according to the experimental observation presented in Fig. 7b, the relationship between the elastic modulus and the temperature is nonlinear. Therefore, a nonlinear form of elastic modulus, $E(\theta)$, is

proposed to describe the temperature dependence:

$$E(\theta) = E_{ref} + E_1\left(\theta - \theta_{ref}\right)^{m_E} \tag{12}$$

where E_{ref} is the elastic modulus at the reference temperature, θ_{ref}, and the terms E_1 and m_E are material constants. The reference temperature (θ_{ref}) in this model is set as the room temperature in Kelvin, 296 K. It should be noted that all temperatures referred to in the model development of this study are absolute temperatures.

For the uniaxial tensile deformation, the von-Mises effective stress form can be derived from the Cauchy stress tensor and expressed as $\sigma_I = 2C_1(\theta)(\lambda_{Ie}^2 - 1/\lambda_{Ie})$, where λ_{Ie} is the intermolecular elastic stretch ratio. According to the definition of the stretch ratio, in the form of effective (true) strain, λ_{Ie} can be expressed as $\lambda_{Ie} = \exp(\varepsilon_{Ie})$, where $\varepsilon_{Ie} = \int \dot{\varepsilon}_{Ie}$ refers to the effective (true) strain of the intermolecular elastic part, carried by the hyperelastic spring, and $\dot{\varepsilon}_{Ie}$ is the corresponding strain rate.

To calculate $\dot{\varepsilon}_{Ie}$, the deformation of each element in the intermolecular branch needs to be identified. There are viscoelastic and viscoplastic parts in the intermolecular branch, and the viscoplastic part comes into effect only when a temperature-dependent stress limit, $\sigma_{lim}(\theta)$, on the friction element is reached. Therefore, the term $\dot{\varepsilon}_{Ie}$ is determined differently at two stages, i.e., before and after the stress limit is reached. In this research, the stress limit is defined as yield stress as shown in Fig. 7a. The JC temperature softening model [5,10,12] is adopted in this work. The stress limit at current temperature is calculated by:

$$\sigma_{lim}(\theta) = \sigma_{ref}\left(1 - \left(\frac{\theta - \theta_{ref}}{\theta_m - \theta_{ref}}\right)^{m_\sigma}\right) \tag{13}$$

where, σ_{ref} is the stress limit at the reference temperature, m_σ is a material constant relating to σ_{lim}, and the term θ_m denotes the melting temperature of a studied material.

Before the stress limit is reached, i.e. $\sigma_I < \sigma_{lim}$, only the spring and the dashpot in the intermolecular viscoelastic part of the intermolecular branch share the total strain. Thus, the intermolecular elastic strain rate is given as:

$$\dot{\varepsilon}_{Ie} = \dot{\varepsilon} - \dot{\varepsilon}_{Iv} \tag{14}$$

where $\dot{\varepsilon}$ is the total effective (true) strain rate, $\dot{\varepsilon}_{Iv}$ is the intermolecular viscous strain rate in the intermolecular viscoelastic part, and $\dot{\varepsilon}_{Iv}$ is represented by a power-law flow rate [34]:

$$\dot{\varepsilon}_{Iv} = \dot{\gamma}_{01}\left(\frac{\dot{\varepsilon}}{\dot{\varepsilon}_{ref}}\right)^{m_{rI}}\left(\frac{\sigma_I}{\eta_{Iv}(\theta)}\right)^{m_{Iv}} \tag{15}$$

where, $\dot{\gamma}_{01}$, m_{rI} and m_{Iv} are material constants controlling the temperature and strain rate sensitivity of the dashpot in the intermolecular viscoelastic part. The term $\dot{\varepsilon}_{ref}$ is the reference strain rate and is defined as the minimum strain rate of a studied strain rate range in this paper. The term $\eta_{Iv}(\theta)$ represents the temperature-dependent coefficient of viscosity of the dashpot in the intermolecular viscoelastic part. With the change in temperature from the glass transition temperature (θ_g) to the melting temperature (θ_m), the material evolves from a solid state to a liquid state, where the coefficient of viscosity decreases gradually from a large number to a small number. Therefore, the expression of $\eta_{Iv}(\theta)$ is shown as:

$$\eta_{Iv}(\theta) = \frac{\eta_{Iv0}}{\left(\frac{\theta - \theta_g}{\theta_m}\right)^{m_{\theta I}}} \tag{16}$$

where, η_{Iv0} and $m_{\theta I}$ are material constants relating to η_{Iv}.

After the stress limit is reached, i.e. $\sigma_I \geq \sigma_{lim}$, both the viscoelastic and viscoplastic parts share the total strain. Therefore, Eq. (14) is rewritten as:

$$\dot{\varepsilon}_{Ie} = \dot{\varepsilon} - \dot{\varepsilon}_{Iv} - \dot{\varepsilon}_{Ip} \tag{17}$$

where $\dot{\varepsilon}_{Ip}$ is the intermolecular plastic strain rate carried by the dashpot in the viscoplastic part, and $\dot{\varepsilon}_{Ip}$ can be expressed as an exponential flow rate according to Refs. [18,34]. In this paper, to address the strain rate and temperature effects on the viscoplastic flow, a revised equation form is proposed, given by:

$$\dot{\varepsilon}_{Ip} = \dot{\varepsilon}_{0p}\left(\frac{\dot{\varepsilon}}{\dot{\varepsilon}_{ref}}\right)^{m_{Ip}}\left\{\exp\left(\frac{1}{C_p(\theta)}\left(\frac{\sigma_I}{\sigma_{lim}} - 1\right)\right) - 1\right\} \tag{18}$$

where $\dot{\varepsilon}_{0p}$ is a viscoplastic multiplier and m_{Ip} is a material constant showing the rate-dependence of the dashpot in the viscoplastic part. The term $C_p(\theta)$ is a temperature-dependent parameter that goes by the following form:

$$C_p(\theta) = \frac{C_{p1}}{\left(\frac{\theta}{\theta_m}\right)^{m_{cp}}} \tag{19}$$

where C_{p1} and m_{cp} are material constants relating to C_p.

2) Network resistance (N)

The stress contributed by the network resistance is defined by a hyperelastic spring which is modelled as an eight-chain model proposed by Arruda and Boyce [13,31]. The eight-chain model form is expressed as:

$$\boldsymbol{\sigma}_N = \frac{C_R(\theta)}{3J_{Ne}}\frac{\bar{\lambda}_L}{\bar{\lambda}_{Ne}}\mathscr{L}^{-1}\left(\frac{\bar{\lambda}_{Ne}}{\bar{\lambda}_L}\right)\left(\boldsymbol{B}_{\boldsymbol{Ne}}^{*} - \left(\bar{\lambda}_{Ne}\right)^2\boldsymbol{I}\right) \tag{20}$$

where, $C_R(\theta)$ is the temperature-dependent modulus of the network resistance, $\boldsymbol{I}$ is the identity tensor, and $J_{Ne} = \det(\boldsymbol{F}_{\boldsymbol{Ne}})$ is the Jacobian determinant associated with the network elastic deformation gradient. The term $\boldsymbol{B}_{\boldsymbol{Ne}}^{*}$ is the deviatoric network elastic left Cauchy-Green deformation tensor, and in the form of $\boldsymbol{B}_{\boldsymbol{Ne}}^{*} = \boldsymbol{F}_{\boldsymbol{Ne}}^{*}(\boldsymbol{F}_{\boldsymbol{Ne}}^{*})^{T}$, and $\boldsymbol{F}_{\boldsymbol{Ne}}^{*} = (J_{Ne})^{-1/3}\boldsymbol{F}_{\boldsymbol{Ne}}$ is the deviatoric part of $\boldsymbol{F}_{\boldsymbol{Ne}}$. The term $\bar{\lambda}_{Ne}$ is the average network elastic stretch ratio, calculated from the form $\bar{\lambda}_{Ne} = \sqrt{\mathrm{tr}(\boldsymbol{B}_{\boldsymbol{Ne}}^{*})/3}$; and in the form of effective (true) strain, $\bar{\lambda}_{Ne}$ can be expressed as $\bar{\lambda}_{Ne} = \sqrt{(\exp(2\varepsilon_{Ne}) + 2\exp(-\varepsilon_{Ne}))/3}$, with ε_{Ne} being the effective (true) strain of network elastic deformation. The last term $(\boldsymbol{B}_{\boldsymbol{Ne}}^{*} - (\bar{\lambda}_{Ne})^2\boldsymbol{I})$ in Eq. (20) could also be expressed in the form of effective (true) strain as $2((\bar{\lambda}_{Ne})^2 - 1/\bar{\lambda}_{Ne})/3$. $\bar{\lambda}_L$ is a material constant associated with the number of "rigid links" between entanglements [13] and the stretching process locks when the number is reached. The term $\mathscr{L}^{-1}(\bar{\lambda}_{Ne}/\bar{\lambda}_L)$ is the inverse of Langevin function and the approximation form is expressed as:

$$\mathscr{L}^{-1}\left(\frac{\bar{\lambda}_{Ne}}{\bar{\lambda}_L}\right) = 3\left(\frac{\bar{\lambda}_{Ne}}{\bar{\lambda}_L}\right) + \frac{9}{5}\left(\frac{\bar{\lambda}_{Ne}}{\bar{\lambda}_L}\right)^3 + \frac{297}{175}\left(\frac{\bar{\lambda}_{Ne}}{\bar{\lambda}_L}\right)^5 + \frac{1539}{875}\left(\frac{\bar{\lambda}_{Ne}}{\bar{\lambda}_L}\right)^7 \tag{21}$$

From the strain energy function of Arruda-Boyce model [31], the eight-chain model could be regressed to the Neo-Hookean model if the higher order infinitesimal is ignored. Therefore, the modulus of the eight-chain model, i.e. the modulus of the network resistance, $C_R(\theta)$, can be inferred from the post-strain hardening modulus as shown by the experimental results in Fig. 8b. Therefore, $C_R(\theta)$ is in a linear relationship with temperature, expressed as:

$$C_R(\theta) = C_{Rref} + C_{r1}\left(\theta - \theta_{ref}\right) \tag{22}$$

where, C_{Rref} and C_{r1} are material constants relating to C_R.

To calculate the deformation carried by the hyperelastic spring, the strain rate of the network elastic deformation can be expressed as:

$$\dot{\varepsilon}_{Ne} = \dot{\varepsilon} - \dot{\varepsilon}_{Nv} \tag{23}$$

where $\dot{\varepsilon}_{Nv}$ is the strain rate of the viscous deformation in the network branch, and can be expressed in a similar form to Eq. (15), shown as:

$$\dot{\varepsilon}_{Nv} = \dot{\gamma}_{02}\left(\frac{\dot{\varepsilon}}{\dot{\varepsilon}_{ref}}\right)^{m_{rN}}\left(\frac{\sigma_n}{\eta_{Nv}(\theta)}\right)^{m_{Nv}} \tag{24}$$

where σ_n is the von-Mises effective stress of the Cauchy stress tensor $\boldsymbol{\sigma}_N$, and $\dot{\gamma}_{02}$, m_{rN} and m_{Nv} are material constants controlling the temperature and strain rate sensitivity of the dashpot in the network resistance. The term $\eta_{Nv}(\theta)$ denotes the coefficient of viscosity of the dashpot in the network resistance, which is also temperature-dependent, and is expressed as:

$$\eta_{Nv}(\theta) = \frac{\eta_{Nv0}}{\left(\frac{\theta-\theta_g}{\theta_m}\right)^{m_{\theta N}}} \tag{25}$$

where η_{Nv0} and $m_{\theta N}$ are material constants relating to η_{Nv}.

3) Heat generation

As has been observed, there is adiabatic heat affecting stress levels during deformation at higher strain rates due to nonelastic deformation. Therefore, it is necessary to correct the total stress with consideration of the thermal effect caused by heat generation. In the form of total effective stress, σ, the total stress after thermal softening correction, σ_c, is calculated by the following form:

$$\sigma_c = \sigma\left(1 - \frac{\theta_c - \theta}{\theta_m - \theta}\right)^{m_h} \tag{26}$$

Table 3
Model summary.

True stress decomposition	$\sigma = \sigma_I + \sigma_N$
The intermolecular resistance (I)	
Viscoelastic part ($\sigma_I < \sigma_{lim}$) $\sigma_{lim}(\theta) = \sigma_{ref}\left(1 - \left(\frac{\theta-\theta_{ref}}{\theta_m-\theta_{ref}}\right)^{m_\sigma}\right)$;	$\sigma_I = 2C_1(\theta)(\lambda_{Ie}^2 - 1/\lambda_{Ie})$ $\lambda_{Ie} = \exp(\varepsilon_{Ie})$ $C_1(\theta) = E(\theta)/(4(1+\nu))$ $E(\theta) = E_{ref} + E_1(\theta-\theta_{ref})^{m_E}$ $\dot{\varepsilon}_{Ie} = \dot{\varepsilon} - \dot{\varepsilon}_{Iv}$ $\dot{\varepsilon}_{Iv} = \dot{\gamma}_{01}\left(\frac{\dot{\varepsilon}}{\dot{\varepsilon}_{ref}}\right)^{m_{rI}}\left(\frac{\sigma_I}{\eta_{Iv}(\theta)}\right)^{m_{Iv}}$ $\eta_{Iv}(\theta) = \frac{\eta_{Iv0}}{\left(\frac{\theta-\theta_g}{\theta_m}\right)^{m_{\theta I}}}$
Viscoplastic part ($\sigma_I \geq \sigma_{lim}$)	$\dot{\varepsilon}_{Ie} = \dot{\varepsilon} - \dot{\varepsilon}_{Iv} - \dot{\varepsilon}_{Ip}$ $\dot{\varepsilon}_{Ip} = \dot{\varepsilon}_{0p}\left(\frac{\dot{\varepsilon}}{\dot{\varepsilon}_{ref}}\right)^{m_{Ip}}\left\{\exp\left(\frac{1}{C_p(\theta)}\left(\frac{\sigma_I}{\sigma_{lim}}-1\right)\right)-1\right\}$ $C_p(\theta) = C_{p1}/\left(\frac{\theta}{\theta_m}\right)^{m_{cp}}$
The network resistance (N)	
Network resistance (N)	$\sigma_N = \frac{C_R(\theta)}{3J_{Ne}}\frac{\bar{\lambda}_L}{\bar{\lambda}_{Ne}}\mathscr{L}^{-1}\left(\frac{\bar{\lambda}_{Ne}}{\bar{\lambda}_L}\right)\left(\frac{2}{3}((\bar{\lambda}_{Ne})^2 - 1/\bar{\lambda}_{Ne})\right)$ $\bar{\lambda}_{Ne} = \sqrt{(\exp(2\varepsilon_{Ne}) + 2\exp(-\varepsilon_{Ne}))/3}$ $\mathscr{L}^{-1}\left(\frac{\bar{\lambda}_{Ne}}{\bar{\lambda}_L}\right) = 3\left(\frac{\bar{\lambda}_{Ne}}{\bar{\lambda}_L}\right) + \frac{9}{5}\left(\frac{\bar{\lambda}_{Ne}}{\bar{\lambda}_L}\right)^3 + \frac{297}{175}\left(\frac{\bar{\lambda}_{Ne}}{\bar{\lambda}_L}\right)^5 + \frac{1539}{875}\left(\frac{\bar{\lambda}_{Ne}}{\bar{\lambda}_L}\right)^7$ $C_R(\theta) = C_{Rref} + C_{r1}(\theta-\theta_{ref})$ $\dot{\varepsilon}_{Ne} = \dot{\varepsilon} - \dot{\varepsilon}_{Nv}$ $\dot{\varepsilon}_{Nv} = \dot{\gamma}_{02}\left(\frac{\dot{\varepsilon}}{\dot{\varepsilon}_{ref}}\right)^{m_{rN}}\left(\frac{\sigma_n}{\eta_{Nv}(\theta)}\right)^{m_{Nv}}$ $\eta_{Nv}(\theta) = \frac{\eta_{Nv0}}{\left(\frac{\theta-\theta_g}{\theta_m}\right)^{m_{\theta N}}}$
Temperature evolution and thermal softening	
	$\sigma_c = \sigma\left(1 - \frac{\theta_c-\theta}{\theta_m-\theta}\right)^{m_h}$ $\theta_c = \theta + \int\dot{\theta}$ $\dot{\theta} = \frac{\omega(\dot{\varepsilon})}{\overline{C}(\theta)}(\sigma_{Iv}\dot{\varepsilon}_{Iv} + \sigma_{Ip}\dot{\varepsilon}_{Ip} + \sigma_{Nv}\dot{\varepsilon}_{Nv})$ $\omega(\dot{\varepsilon}) = \begin{cases} 0 & ;\ \text{if } \dot{\varepsilon} \leq 0.01 \\ \min\left(0.33\left(\log_{10}\left(\frac{\dot{\varepsilon}}{\dot{\varepsilon}_{ref}}\right), 1\right)\right. & ;\ \text{if } \dot{\varepsilon} > 0.01 \end{cases}$ $\overline{C}(\theta) = \overline{C}_{ref} + \overline{C}_1(\theta-\theta_{ref})$

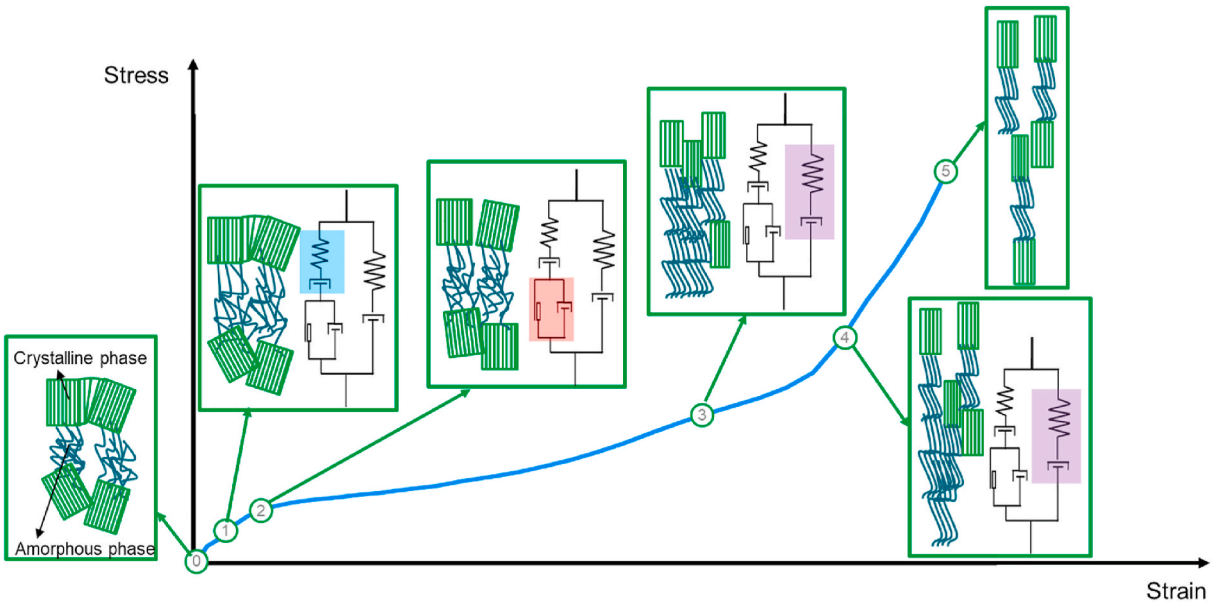


Fig. 12. Schematic illustration of microstructural evolution and the governing parts of the constitutive model, corresponding to the stress-strain response at different stages, during the tensile deformation of semicrystalline thermoplastics [10,38,39].

where m_h is the material constant representing the softening parameter. Term θ_c refers to the corrected temperature when the adiabatic heat comes into effect, and θ_c could be calculated from $\theta_c = \theta + \int \dot{\theta}$, where $\dot{\theta}$ is the temperature evolution during the deformation.

To estimate $\dot{\theta}$, the thermodynamics of the deformation process was explained in detail by Bouvard et al. [33] and Garcia-Gonzalez et al. [5]. In this research, to simplify the equation form, a ω factor is adopted to be a rate-dependent Taylor–Quinney coefficient to measure the ratio of plastic work converting into heat during plastic deformation [35,36]. The thermoelastic coupling behaviour is ignored as the thermal expansion effect is relatively small compared with the intermolecular and network deformation [10]. In addition, to describe the material temperature evolution more accurately, the heat capacity $\overline{C}$ is also revised into a temperature-dependent parameter $\overline{C}(\theta)$ according to the studies of [7,10,37]. The temperature evolution during the deformation can be expressed as:

$$\dot{\theta} = \frac{\omega(\dot{\varepsilon})}{\overline{C}(\theta)}\left(\sigma_{Iv}\dot{\varepsilon}_{Iv} + \sigma_{Ip}\dot{\varepsilon}_{Ip} + \sigma_{Nv}\dot{\varepsilon}_{Nv}\right) \tag{27}$$

where $\sigma_{Iv}, \sigma_{Ip}, \sigma_{Nv}$ are intermolecular viscous stress, intermolecular plastic stress, and network viscous stress, respectively. The term $\omega(\dot{\varepsilon})$ is the ratio of energy that is converted to temperature change to the total plastic dissipation, and this ratio is a function of strain rate. According to the estimation by Garcia-Gonzalez et al. [5], the adiabatic heating effect is anticipated when the strain rate is higher than 0.01 /s. Further, according to the studies of Barba et al. [10], when the strain rate is higher than 1 /s, the deformation is under adiabatic conditions, under which conditions $\omega = 1$. Therefore, $\omega(\dot{\varepsilon})$ of intermediate strain rates (from 0.01 /s to 1 /s) are estimated accordingly. The mathematical form of $\omega(\dot{\varepsilon})$ is defined as below:

$$\omega(\dot{\varepsilon}) = \begin{cases} 0 \quad ; \text{ if } \dot{\varepsilon} \le 0.01 \\ \min\left(0.33 * \left(\log_{10}\left(\dfrac{\dot{\varepsilon}}{\dot{\varepsilon}_{ref}}\right), 1\right)\right); \text{ if } \dot{\varepsilon} > 0.01 \end{cases} \tag{28}$$

The temperature-dependent specific heat capacity $\overline{C}(\theta)$ is expressed as:

$$\overline{C}(\theta) = \overline{C}_{ref} + \overline{C}_1(\theta - \theta_{ref}) \tag{29}$$

where $\overline{C}_{ref}$ is $\overline{C}$ at the reference temperature, and $\overline{C}_1$ is a material constant relating to $\overline{C}$.

The model equations in the form of effective true stress and effective true strain are summarised in Table 3.

4.3. Discussion on the proposed constitutive model and associated deformation mechanisms

This section provides a more detailed explanation on the rationale behind the proposed constitutive model form, and attempts to promote a better understanding on the deformation mechanisms by reflecting on the proposed constitutive model equations.

Fig. 12 schematically illustrates the stress response and microstructural evolution at different deformation stages of semicrystalline thermoplastics under tensile deformation, and correspondingly, the contributing part (i.e. rheological element) of the constitutive model at each stage. Note that, since microstructural investigation is not the focus of this study, the interpretation of the microstructural evolution below is extended discussion, based on literature studies and supported by references.

- Stage 0 shows the initial state of the semicrystalline thermoplastics without load applied. There are well-arranged crystals and amorphous chains connected with each other [21].
- From stage 0 to stage 1, the viscoelastic deformation begins, which is mainly analogous to the viscoelastic part in the intermolecular branch of the constitutive model. During the viscoelastic deformation stage, the amorphous phase is the main deformation phase. The molecular chains in the amorphous region elongate and partially align reversibly. No chain unfolding or fracture in the crystalline region exists at this stage [40]. The stiffness, corresponding to the modulus of the hyperelastic spring in this viscoelastic part, is temperature-dependent during this stage (Eq. (12)), as the mobility of the molecular chains increases when the temperature increases. In the meantime, since there is a certain fraction of free volume [34], molecular relaxation occurs during deformation, shown as the viscous deformation represented by the dashpot in the intermolecular viscoelastic part. At a higher strain rate under the same temperature, there is less time for molecular relaxation, therefore greater stress is required for deformation (Eq. (15)). Further, the coefficient

of viscosity decreases with increasing temperature [34], indicating easier physical movement of the dashpot when the temperature increases (Eq. (16)).

- From stage 1 to stage 2, when the stress limit is reached for activating the friction element in the viscoplastic part, the material yields. Accordingly, the viscoplastic part comes into effect. From a microstructural point of view, the stress limit is the stress barrier for the crystalline phase to overcome to accommodate further deformation, and this barrier is smaller at higher temperatures (Eq. (13)) [34]. After overcoming this stress barrier, the large crystal lamellae separate into smaller crystallites and reorientate towards the tensile axis [40]. The breakage and reorientation of crystallites results in the molecular chains moving more easily and therefore less stress is required [25]. Therefore, the material gradually shifts into a steady flow, shown as temperature- and rate-dependent viscous deformation. In the constitutive model, these dependencies are reflected through the dashpot in the viscoplastic part (Eqs. (18) and (19)).
- From stage 2 to stage 3, the network resistance against stretching starts to gradually take effect, and consequently the strain hardening process begins [41]. Although, in the constitutive model, the network stretch exists in the previous stages, the corresponding stress contribution to the whole system is almost negligible since its magnitude at this stage is relatively small. When the network stretch associated deformation takes the dominant role, the hyperelastic spring in the network branch results in the strain hardening response. In the microstructural aspect, the molecular chains in the amorphous region are reorientated and stretched. Meanwhile, there is transformation from crystallites to amorphous chains, which means the degree of crystallinity changes at the strain hardening stage [3,25]. The molecular mobility at this stage is also temperature-dependent, as the above mentioned behaviours all require less energy at a higher temperature. Therefore, the strain hardening modulus is temperature-dependent (Eq. (22)), as proven by the experimental results. Further, there is also a molecular relaxation process [25], shown as rate-dependent viscous deformation, and the coefficient of viscosity deceases with increasing temperature. As a result, a temperature- and rate-dependent dashpot is also included (Eqs.(24) and (25)).
- From stage 3 to stage 4, the strain hardening continues with the continued deformation, since the amorphous molecular continues to be stretched and the network resistance increases. As the molecular chains are reoriented and stretched reaching the limit of chain extensibility ($\bar{\lambda}_L$ in Eq. (20)) [42], a dramatic upturn in the strain hardening curve occurs. This strain hardening mechanism, from stage 2 onwards, is expressed by the eight-chains model of the hyperelastic spring [31].
- At stage 5, the covalent bonds rupture and the molecular chains slide completely past each other, which results in the fracture of the thermoplastic materials.

4.4. Model calibration

The material constants were identified based on the physical meaning of each rheological element and calibrated based on the experimental test results on PA6 and PEEK. The determination process is explained as follows:

(1) General properties: the material intrinsic properties, including Poisson's ratio ν, glass transition (θ_g) and melting (θ_m) temperatures, material constants relating to the specific heat capacity ($\overline{C}_{ref}$, $\overline{C}_1$), were determined by referencing to existing research papers [7,10,43]. The reference temperature (θ_{ref}) in this paper was the room temperature in Kelvin, 296 K. The modulus at the reference temperature E_{ref} was referenced from the datasheet of the studied materials shown in Table 1.

(2) Material constants that were determined from analysis of experimental results: material constants relating to the elastic modulus, E_1, m_E , were identified according to the experimental results in Fig. 7b. Material constants relating to the stress limit (σ_{ref}, m_σ) were identified by the experimental results in Fig. 7a.

(3) Material constants determined from direct fitting to the stress-strain curves: material constants that show the rate-dependence ($\dot{\gamma}_{01}$, m_{rI}, m_{Iv}, $\dot{\varepsilon}_{0p}$, m_{Ip}, $\dot{\gamma}_{02}$, m_{rN}, m_{Nv}) and material constants that show temperature-dependence (η_{Iv0}, $m_{\theta I}$, C_{p1}, m_{cp}, C_{Rref}, C_{r1}, η_{Nv0}, $m_{\theta N}$),the thermal softening constant m_h that defines the thermal softening behaviour caused by the adiabatic heat, the material constant associated with the number of "rigid links" between entanglements ($\bar{\lambda}_L$), were all determined by fitting the model to the true stress – true strain curves of the experimental

Table 4
Material constants for PA6 and PEEK.

	General properties									
	ν	$\overline{C}_{ref}$ (kJ/m^{3}K)	$\overline{C}_1$	θ_{ref} (K)	θ_g (K)	θ_m (K)	m_h			
PA6	0.48	3.00	0.01	296	320	503	0.5			
PEEK	0.48	2.56	0.01	296	416	616	0.5			
	Material constants associated with the viscoelastic properties									
	E_{ref} (MPa)	E_1 (MPa)	m_E	σ_{ref} (MPa)	m_σ	$\dot{\gamma}_{01}$	m_{rI}	m_{Iv}	η_{Iv0}	$m_{\theta I}$
PA6	2400	−1041.2	0.1554	12.0	3.257	0.001	0.8	0.5	0.25	1.20
PEEK	3200	−1280.1	0.1572	108.0	0.240	0.002	0.8	0.5	3.20	1.26
	Material constants associated with the viscoplastic properties									
	$\dot{\varepsilon}_{0p}$	m_{Ip}	C_{p1}	m_{cp}						
PA6	0.0010	0.5	0.05	3.0						
PEEK	0.0001	0.4	0.04	3.5						
	Material constants associated with the network properties									
	$\dot{\gamma}_{02}$	m_{rN}	m_{Nv}	η_{Nv0}	$m_{\theta N}$	$\bar{\lambda}_L$	$C_{R_{ref}}$ (MPa)	C_{r1} (MPa)		
PA6	0.001	0.1	0.10	490	0.01	2.60	23.958	−0.120		
PEEK	0.001	0.5	1.03	562	0.09	1.22	13.780	−0.045		

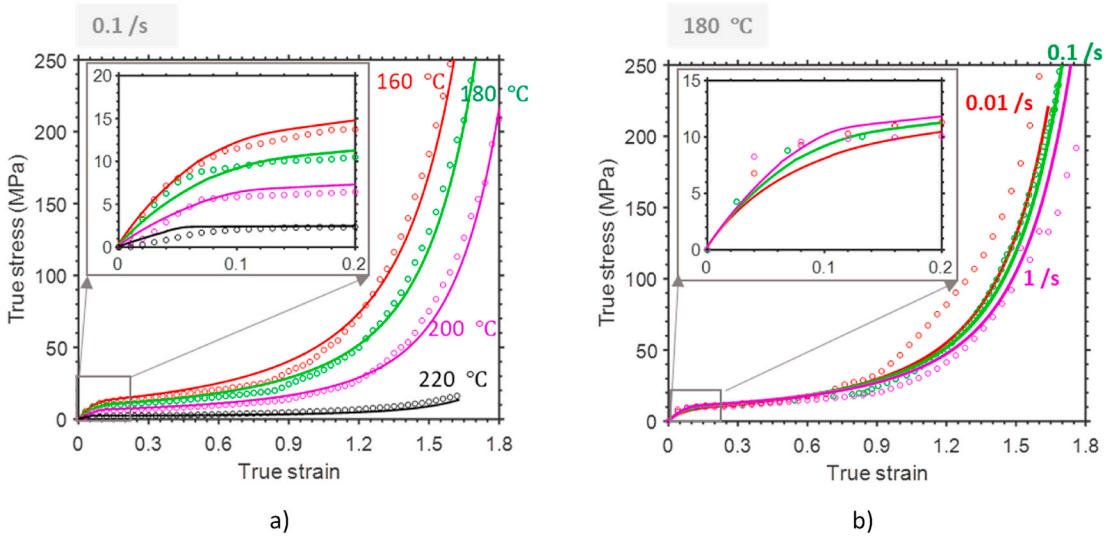


Fig. 13. Comparison between experimental (symbols) and modelling (solid lines) results at different temperatures and strain rates for PA6. a) PA6 at different temperatures (160–220 °C), at a strain rate of 0.1 /s; b) PA6 at different strain rates (0.01–1 /s), at a temperature of 180 °C.

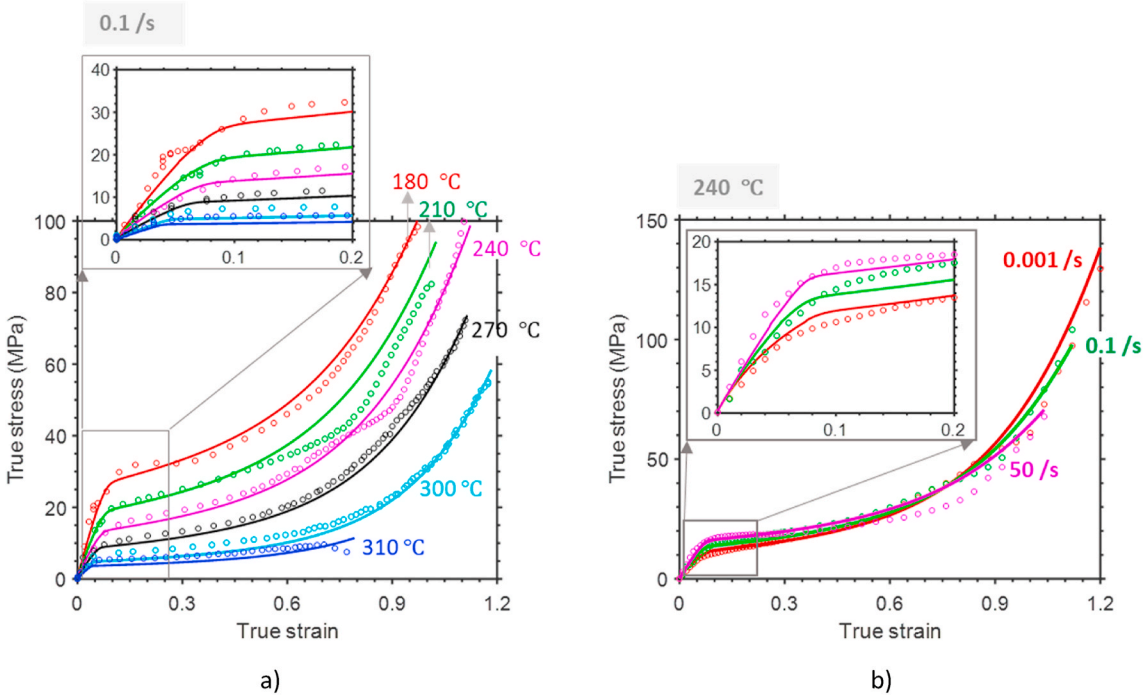


Fig. 14. Comparison between experimental (symbols) and modelling (solid lines) results at different temperatures and strain rates for PEEK. a) PEEK at different temperatures (180–310 °C), at a strain rate of 0.1 /s; b) PEEK at different strain rates (0.001–50 /s), at a temperature of 240 °C.

results. It is noted that the true effective stresses from the experimental results correspond to σ_c in the constitutive model.

The material constants that need to be determined from direct fitting to the stress-strain curves were computed by employing a gradient-based optimisation program. The lsqnonlin algorithm in MATLAB was adopted to solve the nonlinear least-squares problems.

The material constants of the constitutive model for PA6 and PEEK materials are listed in Table 4.

The comparison between the computed and experimental true stress – true strain curves, for PA6 and PEEK, are shown in Fig. 13 and Fig. 14, respectively. As can be seen, the model can accurately capture the deformation behaviour of PA6 and PEEK at the studied temperatures and strain rates. Specifically, Figs. 13a and 14a show the precise prediction of stress responses at the temperature approaching T_m (i.e. 220 °C for PA6 and 310 °C for PEEK), at which the dramatic strain hardening behaviour disappears. Additionally, Figs. 13b and 14b show the model's capabilities in capturing the changing trend of strain rate effects throughout deformation, by taking into account the adiabatic heat influence, which is correlated with the findings in section 3.3.

5. Conclusions

In this research, the thermomechanical behaviour of PA6 and PEEK under solid-state stamp forming conditions has been experimentally studied, and a new physically-based viscoelastic-viscoplastic constitu-

tive model has been developed. The main conclusions drawn from this study are as follows:

1) Temperature has significant effects on the thermomechanical behaviour of the thermoplastics in the studied temperature range (160–220 °C for PA6 and 180–310 °C for PEEK, in both cases at a strain rate of 0.1 /s). The elastic moduli, yield stresses, pre-strain hardening, and post-strain hardening moduli are all found to monotonically decrease with increasing temperature. At temperatures approaching the melting temperature, T_m , of the studied thermoplastics (220 °C for PA6 and 310 °C for PEEK), the post-strain hardening behaviour disappears.
2) Strain rate effects on the tensile properties of the studied thermoplastics are not obvious under the studied range (0.01–1 /s for PA6 and 0.001–50 /s for PEEK, at temperatures between respective T_g and T_m). However, it is worth mentioning that adiabatic heat is found to be a non-negligible factor, which would cause material thermal softening during high-rate deformation.
3) The new physically-based viscoelastic-viscoplastic constitutive model correlates with the deformation mechanisms of the thermoplastics associated with microstructural evolution, by combining the intermolecular and network responses of the semicrystalline thermoplastics. The comparison between the experimental and computed results confirms that this constitutive model is capable of providing an accurate prediction of the viscoelastic-viscoplastic behaviours of PA6 and PEEK under solid-state stamping conditions, by taking into account the temperature and strain rate effects, as well as the effect of adiabatic heat.

Based on the experimental and constitutive modelling studies on the thermomechanical behaviour of solid-state thermoplastics, a guideline on identifying the stamp forming processing windows can be established. With regard to temperature, for stamp forming of pure thermoplastics and discontinuous fibre reinforced thermoplastic polymer matrix composites (TPMCs), temperature ranges of 180–200 °C and 240–300 °C are recommended for PA6 and PEEK, respectively, by considering the elongation and strain hardening characteristics of the materials and the energy cost of manufacturing. For TPMCs reinforced with continuous woven fibres, the large deformation is attributed to the fibre reorientation and fibre bending, and there is more local shear deformation in the polymer constituent. This means strain hardening, which is a key characteristic to enhance drawability [44] in stretch dominated deformation, is no longer the most important consideration. Instead, minimum shear resistance is preferred to guarantee the fibre motions within polymers. Therefore, processing temperatures of $\geq$220 °C and $\geq$310 °C are suggested for composites with PA6 and PEEK matrices, respectively. With regard to the strain rate, since there are no obvious differences in material responses with the varying strain rates in the studied range, high forming speeds within the relevant machine capacities may be adopted to increase productivity.

CRediT authorship contribution statement

Hongyan Wang: Conceptualization, Methodology, Data curation, Formal analysis, Visualization, Writing – original draft. **Haibao Liu:** Supervision, Methodology, Formal analysis, Writing – review & editing. **Zerong Ding:** Formal analysis, Writing – review & editing. **Nan Li:** Investigation, Supervision, Conceptualization, Methodology, Formal analysis, Writing – review & editing, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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